

Causal Inference and Sensitivity Analysis

Matej Havelka

23 March 2026

Pattern Recognition and Bioinformatics — TU Delft

Introduction to Causal Inference

Why Conditional Exchangeability Is Hard

Sensitivity Analysis

Informal Benchmarking

Research Questions and Next Steps

Introduction to Causal Inference

A Simple Causal Question

Smoking

Lung Cancer

A Simple Causal Question



A Simple Causal Question



How would you prove this claim?

How Would You Prove It?

- **Chemistry study:** trace how cigarette chemicals cause cancer.
(Long, indirect, expensive.)

How Would You Prove It?

- **Chemistry study:** trace how cigarette chemicals cause cancer.
(Long, indirect, expensive.)
- **Randomised trial:** randomly assign people to smoke or not, measure lung cancer after 20 years.

How Would You Prove It?

- **Chemistry study:** trace how cigarette chemicals cause cancer.
(Long, indirect, expensive.)
- **Randomised trial:** randomly assign people to smoke or not, measure lung cancer after 20 years.

! NOT ETHICAL — you cannot force people to smoke!

How Would You Prove It?

- **Chemistry study:** trace how cigarette chemicals cause cancer.
(Long, indirect, expensive.)
- **Randomised trial:** randomly assign people to smoke or not, measure lung cancer after 20 years.

! NOT ETHICAL — you cannot force people to smoke!

- **Observational data:** use hospital records, compare smokers vs. non-smokers:

$$\mathbb{E}[\text{Lung Cancer} \mid \text{Smoking} = 1] \quad \text{vs.} \quad \mathbb{E}[\text{Lung Cancer} \mid \text{Smoking} = 0]$$

But Wait — What If There Is a Confounder?

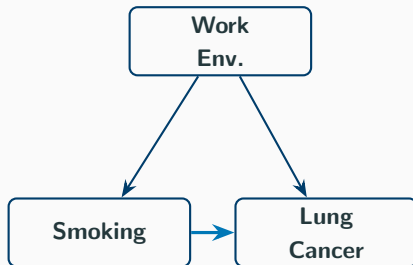
A “what if” scenario:

- Stressful jobs → more smoking.
- Stressful jobs also independently raise cancer risk.

Our raw comparison is polluted:

$$\mathbb{E}[\text{Lung Cancer} \mid \text{Smoking} = 1] \\ \neq \text{effect of smoking alone}$$

It also captures the work environment effect —
we cannot tell them apart.

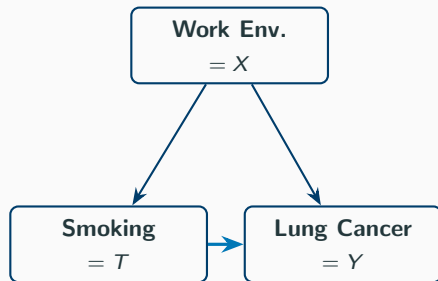


Introducing Causal Notation

We rename the variables:

- **Smoking** $\rightarrow$ Treatment T
- **Work Env.** $\rightarrow$ Confounder X
- **Lung Cancer** $\rightarrow$ Outcome Y
- **$P(\text{Smoking}=1|\text{Work Env.})$** $\rightarrow$
Propensity score $e(X) = P(T = 1|X)$

This abstraction applies to *any* observational study.

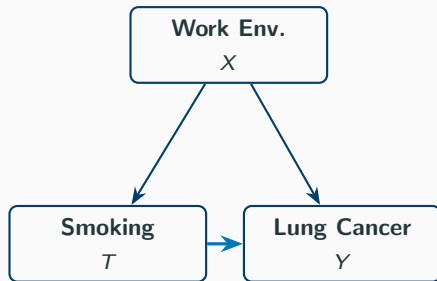


The Core Question of Causal Inference

We can estimate a *conditional expectation* from data:

$$\mathbb{E}[Y \mid T=t] = \int_x y P(y|x, t) \textcolor{red}{P}(t|x) P(x)$$

But this mixes the effect of T with the influence of X on who receives treatment.



The Core Question of Causal Inference

We can estimate a *conditional expectation* from data:

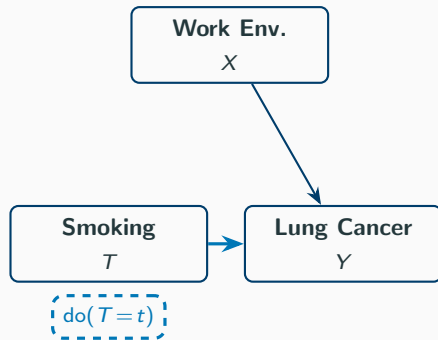
$$\mathbb{E}[Y \mid T=t] = \int_x y P(y|x, t) \mathbf{P}(t|x) P(x)$$

But this mixes the effect of T with the influence of X on who receives treatment.

Causal inference asks: what if we **intervene** and set $T = t$ directly, breaking the link $X \rightarrow T$?

$$\mathbb{E}[Y \mid \text{do}(T=t)]$$

This is what a randomised trial gives us, but done with observational data.



The arrow $X \rightarrow T$ is cut, treatment is assigned regardless of work environment.

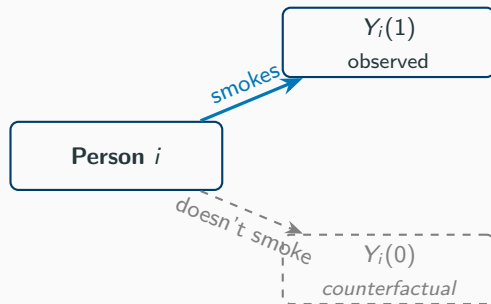
Potential Outcomes

For every person i , imagine two parallel worlds:

- $Y_i(0)$ outcome *if never treated*
- $Y_i(1)$ outcome *if treated*

We only ever observe one.

The other is the *counterfactual* — the road not taken.



Query	Formula	Question asked
ATE	$\mathbb{E}[Y(1) - Y(0)]$	Does smoking affect the <i>population</i> ?
ATT	$\mathbb{E}[Y(1) - Y(0) \mid T=1]$	How much did smoking harm <i>smokers</i> ?
CATE	$\mathbb{E}[Y(1) - Y(0) \mid X=x]$	Effect for people with covariates x ?

All three are impossible to compute directly —
we never observe both $Y_i(0)$ and $Y_i(1)$ for the same person.

From Potential Outcomes to a Regression Problem

Under our identifying assumptions, counterfactuals can be estimated from data. For any unit with covariates x :

$$\underbrace{\mathbb{E}[Y(1) - Y(0) \mid X=x]}_{\text{CATE at } x} = \underbrace{\mathbb{E}[Y \mid T=1, X=x]}_{\mu_1(x)} - \underbrace{\mathbb{E}[Y \mid T=0, X=x]}_{\mu_0(x)}$$

Control model $\hat{\mu}_0$

Train any regression on the **control** subset:

$$\{(x_i, y_i) : T_i = 0\}$$

Treatment model $\hat{\mu}_1$

Train any regression on the **treated** subset:

$$\{(x_i, y_i) : T_i = 1\}$$

$$\widehat{\text{CATE}}(x) = \hat{\mu}_1(x) - \hat{\mu}_0(x) \quad \text{plug in any ML model you like.}$$

Identifying Assumptions

Consistency $T=t \Rightarrow Y = Y(t)$

What we observe matches the potential outcome: a smoker's outcome *is* $Y(1)$.

Identifying Assumptions

Consistency $T=t \Rightarrow Y = Y(t)$

What we observe matches the potential outcome: a smoker's outcome *is* $Y(1)$.

Positivity $0 < \mathbb{P}(T=1 \mid X) < 1$

Every subgroup has *some* treated and *some* control units.

Identifying Assumptions

Consistency $T=t \Rightarrow Y = Y(t)$

What we observe matches the potential outcome: a smoker's outcome *is* $Y(1)$.

Positivity $0 < \mathbb{P}(T=1 \mid X) < 1$

Every subgroup has *some* treated and *some* control units.

Conditional Exchangeability $Y(t) \perp\!\!\!\perp T \mid X$

No *hidden* confounders affect both treatment and outcome.

Why Conditional Exchangeability Is Hard

Is Conditional Exchangeability Realistic?

- **Data is sparse and hard to collect.** Medical records are incomplete, expensive, and rare subgroups may have near-zero overlap between treated and control.

Is Conditional Exchangeability Realistic?

- **Data is sparse and hard to collect.** Medical records are incomplete, expensive, and rare subgroups may have near-zero overlap between treated and control.
- **Some confounders cannot be observed.** Stress, genetics, socioeconomic background, real effects on both treatment and outcome, yet absent from any dataset.

Is Conditional Exchangeability Realistic?

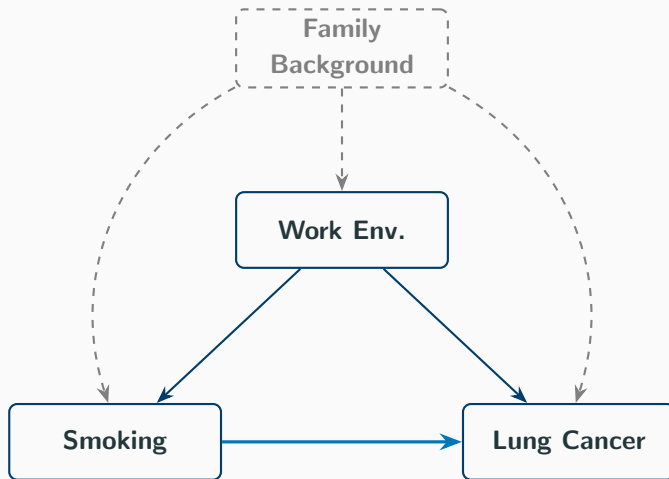
- **Data is sparse and hard to collect.** Medical records are incomplete, expensive, and rare subgroups may have near-zero overlap between treated and control.
- **Some confounders cannot be observed.** Stress, genetics, socioeconomic background, real effects on both treatment and outcome, yet absent from any dataset.
- **Collecting some confounders is unethical.** Race, religion, immigration status, serious ethical and legal concerns.

Is Conditional Exchangeability Realistic?

- **Data is sparse and hard to collect.** Medical records are incomplete, expensive, and rare subgroups may have near-zero overlap between treated and control.
- **Some confounders cannot be observed.** Stress, genetics, socioeconomic background, real effects on both treatment and outcome, yet absent from any dataset.
- **Collecting some confounders is unethical.** Race, religion, immigration status, serious ethical and legal concerns.

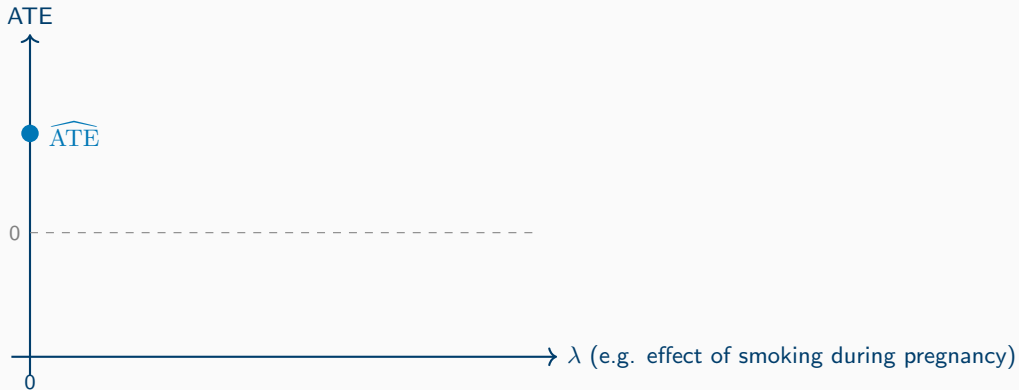
How can we trust observational causal inference
if this assumption is so hard to satisfy?

How Bias Arises



Sensitivity Analysis

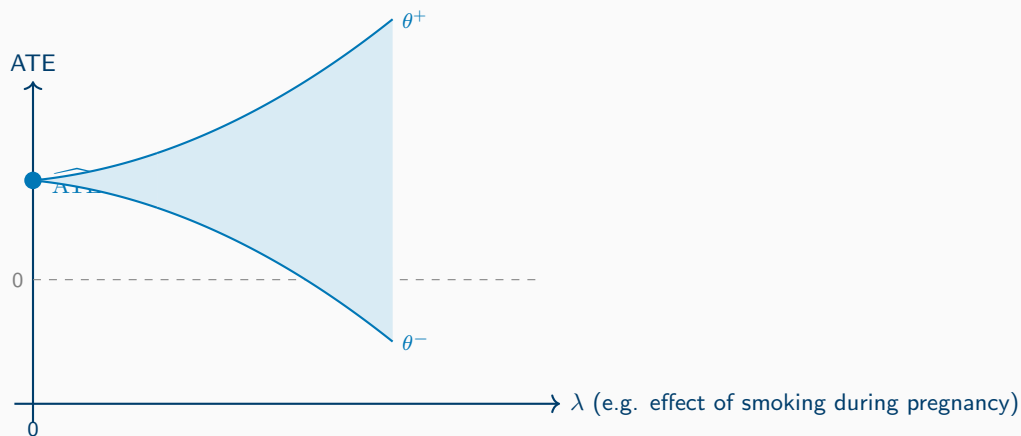
What Is Sensitivity Analysis?



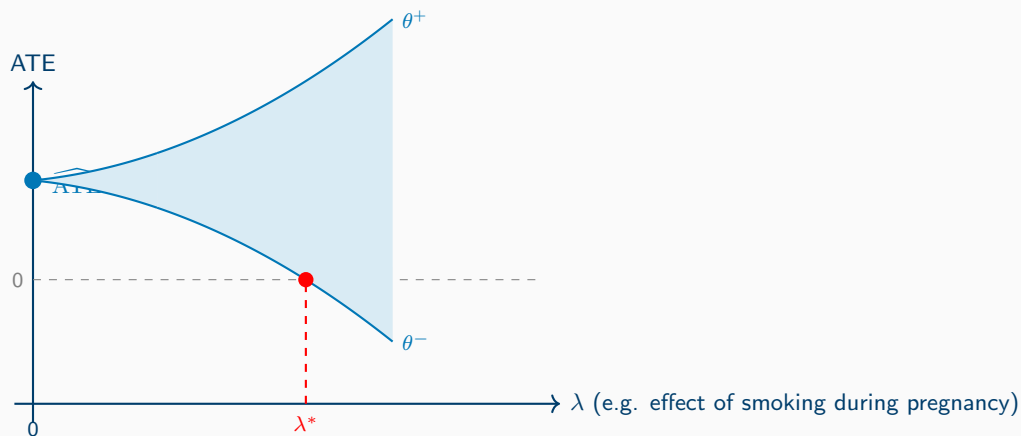
What Is Sensitivity Analysis?



What Is Sensitivity Analysis?



What Is Sensitivity Analysis?



Marginal Sensitivity Model: Setup

Naive idea: bound the hidden propensity as a multiple of the observed one:

$$e(x, u) \leq \Gamma \cdot e(x)$$

Marginal Sensitivity Model: Setup

Naive idea: bound the hidden propensity as a multiple of the observed one:

$$e(x, u) \leq \Gamma \cdot e(x)$$

Problem: $e(x, u)$ is a probability $\in [0, 1]$.
For large $e(x)$, $\Gamma \cdot e(x) > 1$ — impossible!

Marginal Sensitivity Model: Setup

Naive idea: bound the hidden propensity as a multiple of the observed one:

$$e(x, u) \leq \Gamma \cdot e(x)$$

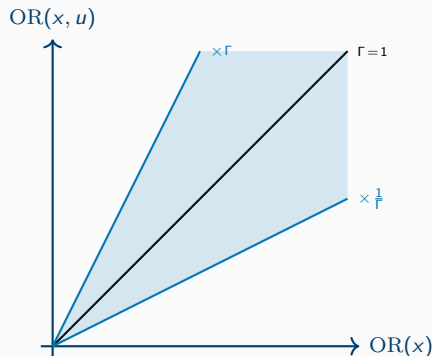
Problem: $e(x, u)$ is a probability $\in [0, 1]$.
For large $e(x)$, $\Gamma \cdot e(x) > 1$ — impossible!

Fix: bound the *odds ratios* instead.

$$\text{OR}(x) = \frac{e(x)}{1 - e(x)}$$

MSM constraint

$$\frac{1}{\Gamma} \cdot \text{OR}(x) \leq \text{OR}(x, u) \leq \Gamma \cdot \text{OR}(x)$$



Marginal Sensitivity Model: Solution

Goal: bound the ATE under the MSM constraint.

Estimator: IPW (inverse propensity weighting):

$$\widehat{\text{ATE}} = \frac{1}{n} \sum_{i=1}^n \left(\frac{t_i y_i}{e(x_i)} - \frac{(1 - t_i) y_i}{1 - e(x_i)} \right)$$

Marginal Sensitivity Model: Solution

Goal: bound the ATE under the MSM constraint.

Estimator: IPW (inverse propensity weighting):

$$\widehat{\text{ATE}} = \frac{1}{n} \sum_{i=1}^n \left(\frac{t_i y_i}{e(x_i)} - \frac{(1 - t_i) y_i}{1 - e(x_i)} \right)$$

Under hidden confounding, the true propensity $e(x_i, u_i)$ differs from $e(x_i)$.

Let $\tilde{e}_i = e(x_i) \lambda_i$ absorb the deviation.

Worst-case optimisation

$$\min_{\{\lambda_i\}} \frac{1}{n} \sum_i \left(\frac{t_i y_i}{\tilde{e}_i} - \frac{(1 - t_i) y_i}{1 - \tilde{e}_i} \right)$$

subject to, for all i :

$$\tilde{e}_i \in \left[\frac{\text{OR}(x_i)}{\text{OR}(x_i) + \Gamma}, \frac{\Gamma \cdot \text{OR}(x_i)}{\Gamma \cdot \text{OR}(x_i) + 1} \right]$$

$$\frac{1}{n} \sum_i \tilde{e}_i = \frac{1}{n} \sum_i e(x_i)$$

Solving gives the **lower bound** $\theta^-(\Gamma)$;

flipping the sign gives $\theta^+(\Gamma)$.

How do I choose Γ ?

Marginal Sensitivity Model: Problem

How do I choose Γ ?

How do I know if a given Γ is realistic?

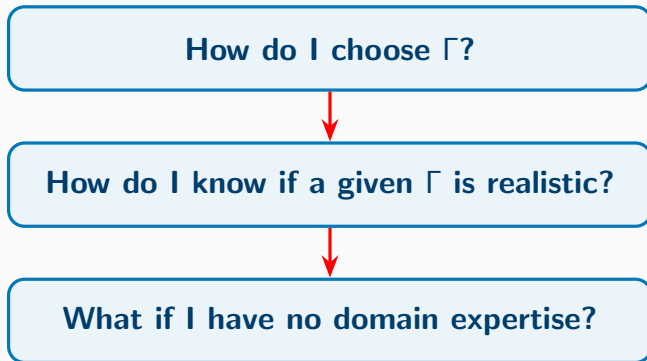
Marginal Sensitivity Model: Problem

How do I choose Γ ?

How do I know if a given Γ is realistic?

What if I have no domain expertise?

Marginal Sensitivity Model: Problem



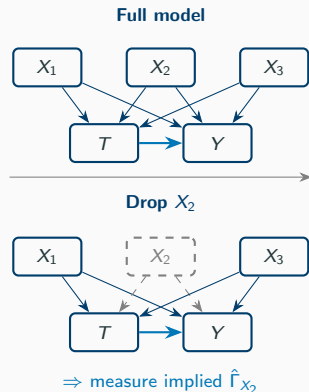
We need a principled, data-driven way to set Γ .

Informal Benchmarking

Assumption: hidden confounding $\leq$ strongest
observed confounder

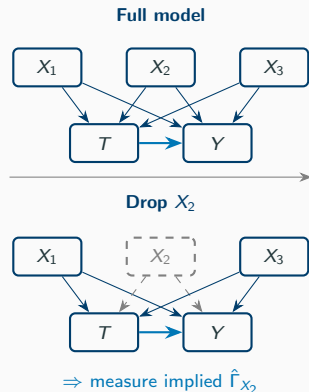
Assumption: hidden confounding $\leq$ strongest observed confounder

- Use each X_j as a *stand-in* for a hidden U
- Drop X_j , measure implied $\hat{\Gamma}_j$



Assumption: hidden confounding $\leq$ strongest observed confounder

- Use each X_j as a *stand-in* for a hidden U
- Drop X_j , measure implied $\hat{\Gamma}_j$
- $\hat{\Gamma}_{\max} = \max_j \hat{\Gamma}_j$ — *data-driven* bound on Γ



Benchmarking procedure

Input: data (X, T, Y) ; fitted $\hat{e}(x)$; sensitivity model.

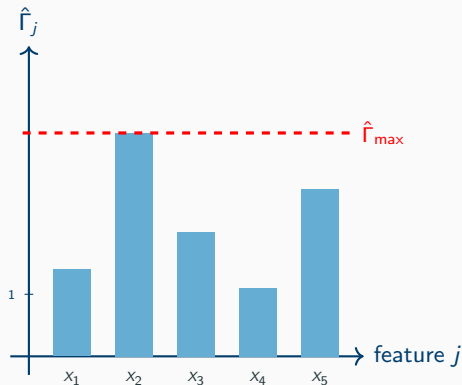
For each X_j :

- Refit $\hat{e}^{(-j)}(x)$ without X_j
- Compute implied $\hat{\Gamma}_j = \max_i \frac{\text{OR}^{(-j)}(x_i)}{\text{OR}(x_i)}$

Output:

$$\hat{\Gamma}_{\max} = \max_j \hat{\Gamma}_j$$

- Bounds hold at $\hat{\Gamma}_{\max} \Rightarrow$ robust to observed-level confounding



$\Gamma \leq \hat{\Gamma}_{\max} \Rightarrow$ no worse than observed confounding.

Research Questions and Next Steps

Research Questions

1. How does the result of informal benchmarking depend on the number of samples?
2. Can we apply the idea of informal benchmarking on the f -sensitivity model?
3. Can we apply the idea of informal benchmarking on the omitted variables bias model?
4. Does extending the informal benchmarking to drop subsets of features lead to better results?
5. What is the interaction between the parameter from informal benchmarking and the bound from sensitivity analysis if the propensity model is incorrect?
6. Can we extend informal benchmarking with PCA? Are there any benefits?

Upcoming weeks

1. Set weekly meeting time throughout Q4
2. Read and understand **Sensitivity Analysis to Unobserved Confounders: A Comparative Review to Estimate Confounding Strength in Sensitivity Models**
3. Research plan due to end of week 1
4. Communication: Email